

AP CSP | Lesson 12: Nested Decisions

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Complete all 12 original questions (30 marks, about 40 minutes). Q5 requires exactly TWO choices. Trace statements in their execution order. Each program statement and each brace occupies its own line. Show reasoning and use the stated valid input range.

1. For IF(age < 12) with age equal to 12, which statement is correct? [1 mark]

A. The condition is true. B. Both IF and ELSE blocks execute.
C. The condition is false, so ELSE is chosen. D. The program cannot compare equal values.

Answer: _____

2. A complete inner IF is inside the ELSE block of an outer IF. When the outer condition is true, what happens to the inner IF? [1 mark]

A. It is skipped. B. It is evaluated as false.
C. It executes after the outer true block. D. It changes the outer result.

Answer: _____

3. The toy price is 5 for ages under 12, 12 for ages 12-64, and 7 for ages 65-120. A nested program first tests age < 12, then tests age < 65 inside ELSE. What fee is selected for age 65? [1 mark]

A. 5 B. 12 C. 0 D. 7

Answer: _____

4. A child price should apply only when age < 12, but the code uses age <= 12. Which input distinguishes these two conditions? [1 mark]

A. 11 B. 12 C. 13 D. 65

Answer: _____

5. Which TWO statements are correct? Select exactly TWO. [1 mark]

A. Every inner condition is evaluated on every run. B. A nested condition is inside another conditional block.
C. An ELSE belongs to all earlier IF statements. D. Reaching the outer ELSE tells us that its outer condition was false.

Answer: _____

6. A program tests age >= 12 first and immediately assigns the adult price on its true path. Its senior-age test is inside that IF's ELSE. What happens for age 70? [1 mark]

A. The senior test selects 7. B. Both prices are selected.
C. The adult branch is chosen and the senior test is skipped. D. Neither condition can be tested.

Answer: _____

Written responses | Questions 7-8

[4 marks]

(b) For age 65, give the inner condition result and selected fee.

[4 marks]

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Written responses | Questions 9-10

9.

[4 marks]

A price program uses the ranges under 12, 12-64, and 65-120 with fees 5, 12, and 7. Give the nearest legal integer immediately below each boundary and the value exactly at each boundary, with their expected fees.

10.

[4 marks]

A program first tests $\text{age} \geq 12$ and assigns fee 12 if true. Only inside ELSE does it test $\text{age} \geq 65$ to assign fee 7; its final ELSE assigns fee 5.

- (a) Give a valid senior-age counterexample with the actual and required fee.
- (b) Describe a correct order of conditions.
- (c) Explain why that order makes the senior case reachable.

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Written responses | Questions 11-12

11. [4 marks]

A club uses this rule: if isMember is true, it checks age < 12 and charges 4 when true or 8 otherwise. If isMember is false, it charges 12 without checking age.

- (a) Give the fee for a member aged 12.
- (b) Give the fee for a nonmember aged 8.
- (c) Is the age test evaluated for the nonmember?
- (d) Which condition controls the ELSE that charges 12?

12. [4 marks]

The original program charges 5 for age < 12, then 12 for age < 65 in the outer ELSE, otherwise 7. Change it so children are under 13, seniors begin at 60, and the senior fee is 6. Adult and child prices stay unchanged.

- (a) Write the revised outer condition.
- (b) Write the revised inner condition.
- (c) State the revised senior assignment.
- (d) Give expected fees for ages 12 and 60.
